

BeaverWorld: Decolonizing Climate Education through Place-Based Interactive Simulation

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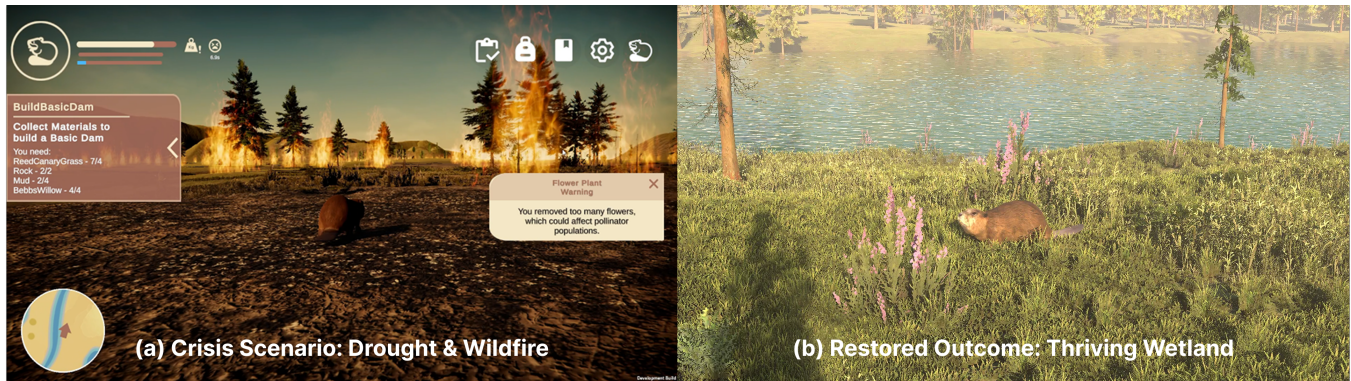


Figure 1: (a) Drought-stricken landscape with procedural wildfires. (b) Restored wetland after strategic dam construction by the player.

Abstract

Climate change education often struggles with abstract concepts and narratives that disengage students. This paper introduces *Beaver World*, a place-based 3D serious game set in the Okanagan region of British Columbia. Drawing on Indigenous ecological perspectives, the project integrates real-time environmental simulation to teach water sustainability. *Beaver World* utilizes a survival-strategy genre where players embody a beaver, a keystone species, to actively restore a degraded watershed. By transforming simple game mechanics into complex ecological feedback loops (e.g., dam building to mitigate drought), the game provides a novel framework for experiential STEAM education. We discuss the assignment design,

scalability for different learner levels, and the toolkit provided to instructors to facilitate adoption.

CCS Concepts

- **Applied computing** → **Interactive learning environments**;
- **Social and professional topics** → *Computing education programs*;
- **Human-centered computing** → *Interactive systems and tools*.

Keywords

Serious Games, Indigenous Knowledge, Environmental Simulation, Place-Based Learning, Water Sustainability, Interactive Storytelling

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1 Introduction

This research takes place on the unceded, ancestral territory of the Syilx (Okanagan) people. The Syilx have sustained a deep and reciprocal relationship with *tmx^wúla?x^w*, the land, and *siw^wkk^w*, the water, cultivating systems of ecological governance that have shaped the Okanagan watershed for millennia [Okanagan Nation Alliance 2014, 2022; Sam 2008, 2013; Sam and Armstrong 2013].

Education is widely recognized as a critical strategy for addressing the challenges posed by climate change [Meya and Eisenack 2018; Wu and Lee 2015; Xu et al. 2026a]. This is reflected in initiatives such as the United Nations Decade of Education for Sustainable Development. Moreover, the Canadian government has recently invested \$2.9 million in organizations engaged in environmental education [Environment and Climate Change Canada 2024].

Yet, educators frequently lack the resources and training needed to teach climate-related topics [Goodchild et al. 2017; Schatz 2021].

Beaver World addresses this gap by building on the pioneering work of Sam [Sam 2008, 2013; Sam and Armstrong 2013], whose scholarship on Syilx water governance laid the groundwork for this project. Guided by the Syilx Water Strategy [Okanagan Nation Alliance 2022] and broader Indigenous perspectives on water and ecological thinking [Simpson 2025; Yunkaporta 2019], the project draws on two formative predecessor initiatives: the *Waterways* exhibition [Correia et al. 2024; Dulic et al. 2023] and the *Okanagan Water Futures* project [Correia et al. 2025], as well as Future Delta 2.0 [Angel et al. 2015]. Set in the Okanagan landscape, a region facing severe water security issues [Neilsen et al. 2018], the game engages students from a cross-cultural perspective applying Indigenous values on water [Okanagan Nation Alliance 2014, 2022]. This paper presents *Beaver World* as an Engaging Education Technique and Assignment (EETA) that uses computer graphics to visualize invisible ecological systems.

2 EETA Metadata

To ensure broad applicability across diverse educational settings, the assignment is structured as follows:

Table 1: Assignment Metadata

Field	Description
Summary	A scaffolded 3D simulation where students act as beavers to restore a drought-stricken watershed.
Outcomes	Systems Thinking (cause-effect loops); Stewardship (ecological reciprocity); Data Literacy (graphical feedback).
Audience	Undergraduate Environmental Science; adaptable for Grades 10–12. Standalone executable, no coding required.
Deliverables	Game Executable; Facilitation Guide; Reflection Worksheets; Assessment Rubrics.

3 The Assignment Design: Fun & Scalability

Beaver World transforms complex climate data into a playful, scaffolded assignment. The core philosophy is Playful Stewardship: deep learning wrapped in the engaging mechanic of becoming a beaver, supported by research on educational games for climate science [Besalti and Smith 2024; Smith et al. 2019], meaningful mechanics [Molnar and Kostkova 2013], and challenge calibration [Sun et al. 2026]. Using genre conventions from survival games like *Timberborn* [Mechanistry 2021] and *Alba* [ustwogames 2020], the assignment offers three levels: (1) **Observation**, where students explore the 3D environment and use the Field Analysis Tool to capture environmental indicators of degradation; (2) **Intervention**, the core task, where they strategically place dams to raise the water table before the in-game winter, teaching the link between water retention and ecosystem survival; and (3) **System Stress Test**, where advanced students face a procedural Climate Crisis and must optimize biodiversity metrics in the Wetland Tracker to create a resilient firebreak.

The game emphasizes the joy of restoration. When a student successfully places a dam, the graphics engine provides immediate, gratifying feedback: water rushes into dry areas, the terrain transforms from brown to lush green, and ambient audio shifts from wind to bird songs. This positive feedback loop encourages students to experiment with the simulation rather than fear failure.

4 Materials & Implementation

A key goal of this EETA is ease of adoption for instructors who may not be computer graphics experts. The assignment is distributed as a standalone, platform-independent executable requiring no coding from students. To support the instructor, the assignment includes: a Facilitation Guide for introducing the ecological and cultural context of the Okanagan watershed before gameplay; an in-game Wetland Tracker providing real-time biodiversity data (see Figure 2, derived from project data [Correia et al. 2025]); a printable Eco-Log where students track in-game decisions and link actions (e.g., cutting a tree) to consequences (e.g., soil erosion); and assessment rubrics aligned with the learning outcomes. Our next phase involves testing the efficacy of this toolkit in diverse classroom settings to assess how easily educators can integrate it into their STEAM curriculum.

5 Evaluation & Findings

The development followed a User-Centered Design process involving 20 university students. Our initial iteration took the form of a gamified textbook, which saw low retention, consistent with findings that text-heavy games often fail to engage [Molnar and Kostkova 2013]. The redesign shifted to a systemic survival model. Post-play role-play debates demonstrated that students better retained concepts of interconnectedness when they physically manipulated the environment. The visual transformation of the 3D terrain proved to be a more powerful teacher than text, aligning with observations that games can lead to stickier learning [Pfirman et al. 2021]. To validate these qualitative insights, we plan to implement a formal, quantitative evaluation using pre- and post-tests [Xu et al. 2026b] to measure learning outcomes and compare them against traditional climate education methods.

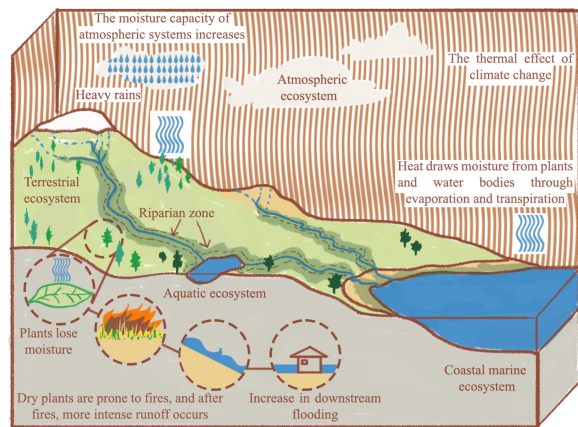


Figure 2: The Ecological Simulation Model. Rising temperatures accelerate evapotranspiration, drying vegetation and increasing wildfire susceptibility. Destabilized soil causes runoff and flooding. Students intervene in this cycle through dam construction.

6 Conclusion

Beaver World demonstrates that effective climate education requires moving beyond didactic methods toward experiential simulations. By combining advanced computer graphics with place-based ecological knowledge and a scalable assignment structure, we provide a tool that allows students to visualize and enact the resilience of natural systems [Correia et al. 2025]. Future work will rigorously test the pedagogical efficacy of this design and explore how the Indigenous knowledge frameworks [Okanagan Nation Alliance 2014, 2022; Sam 2008, 2013; Sam and Armstrong 2013; Simpson 2025; Yunkaporta 2019] embedded in the game may deepen the learning experience.

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